

Shihao Cheng (程世豪)

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I am a Master's student at **Wuhan University**, supervised by Prof. Zhigang Tu. **Expected graduation: June 2027**. Prior to this, I obtained my B.S. degree from **Harbin Institute of Technology** in 2024, ranking **first** in my class.

My research interests lie in multimodal generation, with a current focus on **Video Generation** and **World Model**.

EDUCATION

Wuhan University

M.S. Student. Communication and Information Systems

Wuhan, Hubei
Sep. 2024 – Now

Harbin Institute of Technology

Bachelor of Information Engineering

Harbin, Heilongjiang
Sep. 2020 – Jun. 2024

➤ Avg Score: 92.3/100, Rank: 1/29.

➤ **National Scholarship 2023. ¥8,000 RMB, Top 1%.** Provincial Outstanding Graduate, 2024, Top 1%.

SELECTED PUBLICATIONS

Unison: **Harmonizing** Motion, Speech, and Sound for Human-Centric Audio-Video Generation

First Author | [\[paper\]](#) [\[website\]](#)

European Conference on Computer Vision (ECCV 2026)

- Proposed a harmonization strategy that decouples speech and SFX generation to resolve acoustic interference.
- Designed a bidirectional cross-modal forcing strategy for precise motion-audio synchronization.

InteractiveAvatar: Real-Time Streaming Video Generation for **Consistent** and **Intent-Aware** Avatars

European Conference on Computer Vision (ECCV 2026) | [\[paper\]](#)

- Proposed a Memory mechanism to preserve consistency and prevent temporal drift in infinite-length streaming generation.
- Designed an LLM-driven Reasoning-Reaction Module for intent-aware, real-time user-avatar interaction.

GeoMMBench&GeoMMAgent: Toward **Expert-Level** Multimodal Intelligence in Geoscience&Remote Sensing

Co-First Author | [\[paper\]](#) [\[website\]](#)

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026, Highlight)

- Constructed a rigorous expert-crafted benchmark spanning multiple disciplines, sensor modalities, and task hierarchies, benchmarking 36 MLLMs to expose systematic gaps in domain knowledge, perception, and reasoning.
- Designed a multi-agent framework coordinating retrieval, perception, and reasoning tools for expert-level performance.

OwlSight: A Robust Illumination Adaptation Framework for Dark Video Human Action Recognition

First Author | [\[paper\]](#)

IEEE Transactions on Circuits and Systems for Video Technology, 2025 (T-CSVT, IF: 11.1)

- Proposed an illumination-adaptation framework for robust human action understanding under complex, non-uniform low-light scenes, achieving SOTA across four benchmarks.

EXPERIENCE

Research Intern, Tencent Hunyuan

Apr. 2026 – Now

- Researching real-time streaming multi-shot audio-video generation with consistent cross-modal alignment over long-horizon scene transitions.
- Exploring controllable video and audio joint generation with fine-grained user control.

Research Intern, TeleAI

Aug. 2025 – Apr. 2026

- Advised by: Dr. Shansong Liu
- Built a human-centric audio-video generation model with balanced, synchronized motion, speech, and sound, building the distributed training and data curation pipeline (3M audio-visual clips & 40M audio segments / 110K+ hours), leading to Unison ([ECCV 2026](#)).
- Developed a real-time, infinite-length streaming avatar generation model with intent-aware interaction, leading to InteractiveAvatar ([ECCV 2026](#)), successfully launched on Telestudio.

Research Intern, SenseTime

Mar. 2025 – Jun. 2025

- Advised by: Dr. Jiahao Wang
- Built an audio-visual data curation pipeline and SFT on Qwen2.5-Omni. Applied GRPO with spatio-temporal anchor tracing and modality-contribution adversarial rewards to suppress temporal hallucination and modality bias, improving synchronization and complex-scene reasoning.